

積の微分の証明

関数 $f(x), g(x), h(x)$ が微分可能であるとせよ

$$1. \{f(x)g(x)\}' = f'(x)g(x) + f(x)g'(x)$$

$$2. \{f(x)g(x)h(x)\}' = f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)$$

(1の証明)

$$\begin{aligned} \{f(x)g(x)\}' &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left\{ \frac{f(x+h) - f(x)}{h} g(x+h) + f(x) \frac{g(x+h) - g(x)}{h} \right\} \\ &= f'(x)g(x) + f(x)g'(x) \quad \square \end{aligned}$$

(2の証明)

$$\begin{aligned} \{f(x)g(x)h(x)\}' &= [f(x)\{g(x)h(x)\}]' \\ &= f'(x)\{g(x)h(x)\} + f(x)\{g(x)h(x)\}' \\ &= f'(x)g(x)h(x) + f(x)\{g'(x)h(x) + g(x)h'(x)\} \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x) \quad \square \end{aligned}$$